



BRAC UNIVERSITY
Department of Computer Science Engineering
Mid-Term Examination, Fall 2024
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: 40 (20 x 2)

Time: 60 Minutes

- Total no. of questions are **three (3)**. Answer any **two (2)**.
- Figure in bracket [] next to each question indicates marks for that question.
- Use $V_{\gamma}(\text{diode}) = 0.6 \text{ V}$, $V_D(\text{conducting voltage of diode}) = 0.7 \text{ V}$, $V_{\gamma}(\text{transistor}) = 0.5 \text{ V}$, $V_{BE}(\text{forward active}) = 0.7 \text{ V}$, $V_{BE}(\text{saturation}) = 0.8 \text{ V}$, $V_{CE}(\text{saturation}) = 0.2 \text{ V}$, $\beta_F = 30$, $V_{BC}(\text{reverse active}) = 0.7 \text{ V}$ for all the questions **unless otherwise stated**.

Name:	ID:	Section:
-------	-----	----------

Question 1

[20]

Answer the following questions from the given driver-load circuit. [$V_{OL} = 0.2 \text{ V}$, $V_{OH} = 11 \text{ V}$, $\beta_F = 25$, $\beta_{forced} = 0.1$] [High input = 12 V, Low input = 0.2 V]

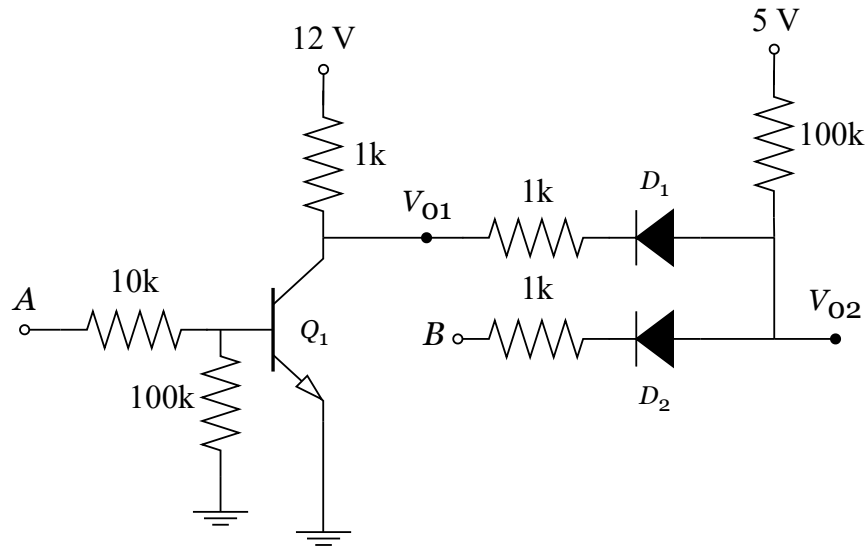


Figure 1

CO2	(a)	Calculate the output voltage V_{O2} for A = Logic 1, B = Logic 1.	[6]
CO2	(b)	Calculate the power dissipation of the driver circuit for A = Logic 0, B = Logic 1.	[6]
CO2	(c)	Calculate maximum fanout. [Hint: Consider B = Logic 1, DL AND gate as loads and RTL NOT gate as driver]	[8]

[20]

For the circuit in Figure 2, assume **logical Low** is **0.2 V**.

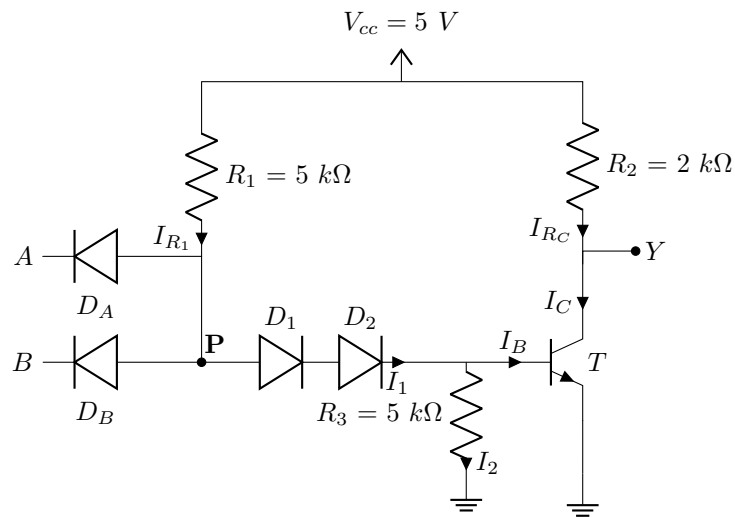


Figure 2

CO2	(a)	Create a truth table and determine which logic gate it represents; you must show your calculation. Also find β_{min} for this circuit.	[10]
CO2	(b)	Calculate the average power dissipation.	[4]
CO2	(b)	Given that the loads are identical to the driver, determine the maximum fanout.	[6]

Question 3

[20]

For the given TTL circuit in Figure 3, $\beta_F = 30$, $V_{BE(sat)} = 0.8V$, $V_{CE(sat)} = 0.2V$, and $\beta_R = 0.1$. Assume **logical High** is **9V** and **logical Low** is **0V**.

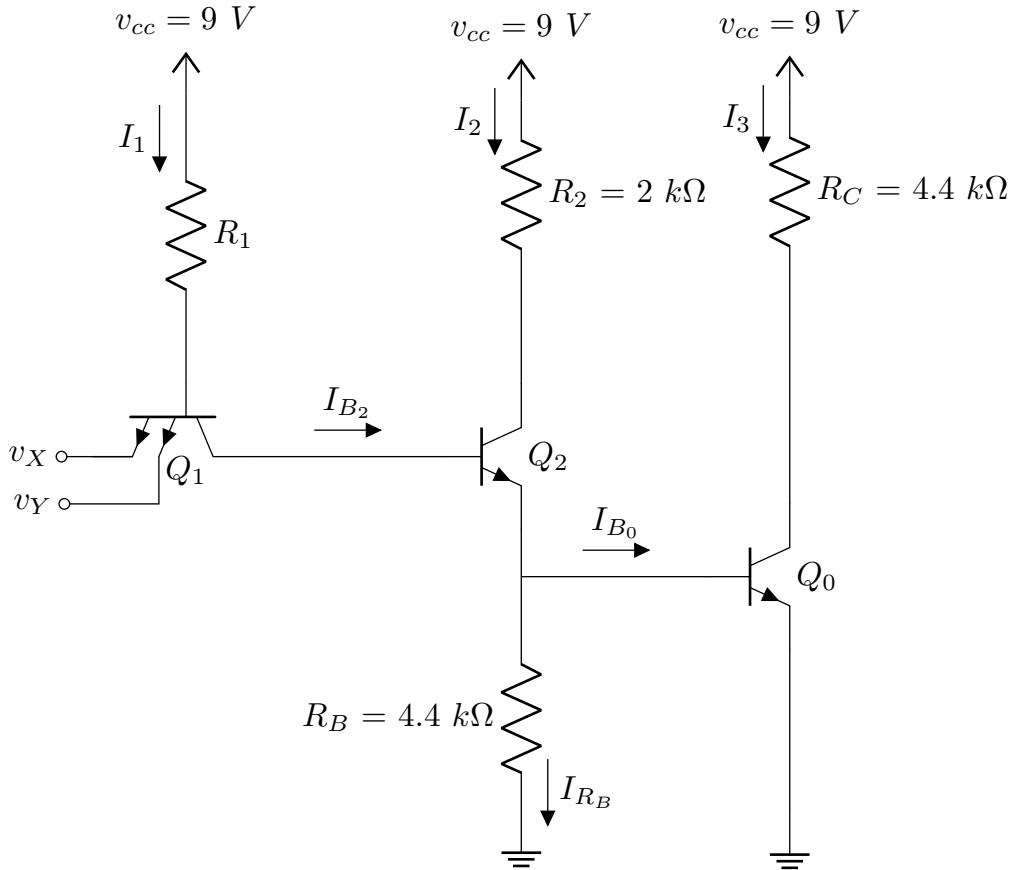


Figure 3

CO1	(a)	Determine the power dissipation of the TTL circuit when both inputs are HIGH. You must show your calculation. Assume $R_1 = 6.7 k\Omega$ for this question only.	[14]
CO1	(b)	Suppose you are selling this TTL logic gate to a customer and he states a strict criteria of keeping the power dissipation below $1.23 mW$ when both inputs are LOW (0 V). Determine the maximum value of R_1 that you can use to satisfy this criteria.	[6]